

Module 5: Pandas for Data Engineering

1. DataFrames

A DataFrame is a two-dimensional table with rows and columns.

Code:

```
import pandas as pd
df = pd.DataFrame({'Name':['John','Sara'],'Salary':[50000,60000]})
```

2. Read CSV

Read data from a CSV file into a DataFrame.

Code:

```
df = pd.read_csv('employees.csv')
```

3. Data Cleaning

Remove duplicates and clean unwanted data.

Code:

```
df = df.drop_duplicates()
df = df.dropna()
```

4. Filtering

Select rows based on a condition.

Code:

```
high_salary = df[df['Salary'] > 50000]
```

5. GroupBy

Group data and calculate summary statistics.

Code:

```
df.groupby('Department')['Salary'].mean()
```

6. Merge

Combine two DataFrames using a common column.

Code:

```
result = pd.merge(df1, df2, on='Emp_ID')
```

7. Missing Values

Identify and replace missing values.

Code:

```
df.isnull().sum()
```

```
df.fillna(0, inplace=True)
```

8. Exploratory Data Analysis (EDA)

Understand the dataset using descriptive statistics.

Code:

```
df.head()
```

```
df.info()
```

```
df.describe()
```

Summary

- Pandas simplifies data manipulation.
- DataFrames are the core structure.
- Read, clean, filter, merge, and analyze data efficiently.
- EDA helps understand data before building ML models.